



Who knows best? Effects of speech disfluencies on incentivized decision-making

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Abstract

Previous work has shown that speech disfluencies can negatively impact judgments about a speaker's competence and confidence. However, these effects have primarily been examined with Likert-type rating scales, which are not informative about how judgments might translate to behavior. Does the presence of disfluencies actually guide decision-making when listeners stand to gain concretely from making the correct choice? We sought to address this question with a web-based decision task in which participants were asked to choose between two conflicting sources of information. Our results suggest that listeners do take speech fluency into account when deciding who or what to believe.

Index Terms: speech perception, decision-making, text-to-speech, paralinguistics

1. Introduction

Spontaneous speech is rich with information about a speaker's knowledge, attitudes and traits, cognitive effort, and the state of the speech planning process. Speech disfluencies (e.g., filled and silent pauses, repetitions, repairs, or lexical fillers) comprise one such source of information that listeners might use to make inferences about a speaker.

The evidence that speech disfluencies affect evaluations of speakers is increasingly robust. In a few cases, disfluencies have been shown to positively impact the way speakers are perceived, for example, making them sound more spontaneous [1] or friendly [2]. By and large, however, listeners seem to rate disfluent speakers more negatively on a variety of dimensions. Disfluent speech is associated with lower ratings of social desirability [3, 4], competence [5, 6], confidence [7, 8, 5, 6], sincerity [6], and trustworthiness [2], as well as a higher degree of anxiety [9].

It is less clear how, or whether, responses on Likert-type rating scales would carry over to actual behavior. The degree and nature of the relationship between judgments and behavioral outcomes is often complex [10], and decision-making can involve heuristics that do not make use of all available information to guide behavior [11]. The fact that hesitant speakers might sound less competent does not guarantee that a listener would decide to discount what they say.

Previous work has connected speech disfluencies to listener behaviors, specifically in regards to deception. Although speech disfluencies do not seem to actually reflect dishonesty, and in fact, the opposite may be true [12], listeners nonetheless seem to think that speakers are more disfluent when lying [13, 14, 15]. Experiments involving bluffing games have shown that when speakers produce more disfluencies, listeners are more likely to

think they are being deceptive, and they adapt their gameplay accordingly [12, 16].

While these studies provide convincing evidence that disfluent speech might affect decisions, bluffing games inherently produce an expectation of potential dishonesty. This makes it unclear whether disfluencies still affect behavior when a listener is not explicitly prompted to infer and perhaps mistrust the speaker's intent. Many decisions about who and what to believe occur in contexts where there is no expectation that someone might deliberately mislead us.

It is also unclear whether listeners take fluency into account when trying to decide *who* to believe, as opposed to *whether* to believe: in other words, when asked to discriminate between two conflicting sources of information. Asking listeners to evaluate competing information sources is tricky to implement in a more naturalistic settings, since even the two party interactions involved in bluffing games make it difficult to disentangle the effects of specific speech behaviors, such as disfluencies, from more global impressions of the speaker.

The present study seeks to directly investigate whether speech disfluencies impact decision-making by presenting listeners with two conflicting sources of information that vary in terms of fluency and incentivizing decision accuracy. We hypothesize that listeners will select fluent answers as correct at a higher rate than disfluent answers.

2. Method

2.1. Materials

Our stimuli consisted of short conversations between two speakers discussing possible answers to trivia questions, synthesized using the Eleven Flash v2.5 text-to-speech model from ElevenLabs¹.

2.1.1. Design criteria

We designed the stimuli with a number of criteria in mind. First, we wanted questions with an objectively correct answer, rather than matters of opinion. However, we also wanted to ensure that participants could not base their decisions on prior knowledge or answers from search engines or chatbots, since we were interested in how the delivery of the answers affected their decisions. We also wanted to rule out, to the greatest extent possible, that our results could be explained by more general impressions of a specific voice. Finally, we wanted to avoid inherent biases in terms of which answers seemed most plausible.

Table 1: Variations of stimulus items for counterbalancing

	Answer order: AB	Answer order: BA
<i>Fluent -disfluent</i>	Speaker A: Oh, I know this one. The 7th leading cause of death in 1900 was liver disease.	Speaker A: Oh, I know this one. The 7th leading cause of death in 1900 was cancer.
	Speaker B: No, I don't think so. I remember the uh, the 7th leading cause of death being uh, cancer.	Speaker B: No, I don't think so. I remember the uh, the 7th leading cause of death being uh, liver disease.
<i>Disfluent - fluent</i>	Speaker A: Oh, I know this one. The uh, the 7th leading cause of death in 1900 was uh, liver disease.	Speaker A: Oh, I know this one. The uh, the 7th leading cause of death in 1900 was uh, cancer.
	Speaker B: No, I don't think so. I remember the 7th leading cause of death being cancer.	Speaker B: No, I don't think so. I remember the 7th leading cause of death being liver disease.

2.1.2. Guesstimation style trivia questions

The trivia questions for our stimuli were inspired by the guesstimation games in [17]. The items we used were based on ranked lists related to topics such as names, weather statistics or causes of death, e.g., “What was the 7th leading cause of death in 1900?” The rankings were gathered from publicly available data, primarily from American government agencies, such as the US National Weather Service, Social Security Administration, Census Bureau, and Center for Disease Control.

2.1.3. Answer alternatives

Next, we selected pairs of possible answers that were incorrect by one rank in either direction (e.g., the answers to “What was the 7th leading cause of death in 1900?” were liver disease, the 6th ranked cause of death, and cancer, the 8th ranked cause). The answers were incorrect, but not unreasonable: they could be thought of as nearly correct answers since nearby items on a list ranked by objective criteria differ quantitatively rather than categorically.

2.1.4. Filtering on ease of finding external answers

We then queried Google and ChatGPT with these questions. If either tool yielded the correct answer, or one of our two nearly correct answers, the item was discarded. It would have been possible, but very time consuming, for participants to find the correct answers themselves by locating and manually searching through statistical databases or scans of historical documents.

2.1.5. Filtering on answer bias

In order to avoid answer bias, we conducted a pilot study in which 20 participants were asked to choose between the two alternatives, counterbalanced for presentation order. A Z test of proportions was used to find answer pairs where each alternative was equally likely to be chosen (i.e., the rate of choosing a given answer did not differ from chance). Eight trivia questions and corresponding answer pairs met this criterion and were used to create test items. An additional 4 were used as filler items.

¹<https://elevenlabs.io/>

2.1.6. Design of the final stimuli

After selecting the trivia questions we synthesized short dialogues in which pairs of speakers discussed the possible answers. Examples of the text used to synthesize the utterances are shown in Table 1. In each exchange, the first speaker stated that they knew the answer to the question, then asserted their answer. Speaker 2 expressed disagreement and provided their own answer. The question itself was implicitly provided within the first speaker's utterance.

One utterance from each pair was disfluent. The same types of disfluencies (a repetition of “the” with the filled pause “uh” and a filled pause before the answer) were used for all disfluent utterances.

2.1.7. Synthesis of the final stimuli

In order to give the speakers distinct but similar voices, we used the ElevenLabs voice design tool, which creates TTS voices based on a text description input by the user. We created four voices with this tool, two male and two female, using the same description aside from speaker gender: “American [man/woman] in [his/her] mid 30s. Conversational, casual, spontaneous speech.”

2.2. Procedure

2.2.1. Variations and balancing

We constructed a total of 16 variations of each utterance pair by varying the speaker order, the gender of the speakers (both male or both female), the order of the answers, and which speaker was disfluent. This allowed us to counteract any potential effects of ordering or characteristics of a specific voice by counterbalancing these variations across participants, while also obfuscating the fluency manipulation. The fluency/answer order variations are shown in Table 1. Statistical tests were subsequently carried out as an additional check for ordering or voice confounds. These are detailed in Section 3.2.

In addition to the test items described above, each participant was presented with the same four filler items in which both speakers were either fluent or disfluent. The fillers were intended to further conceal the fluency manipulation by includ-

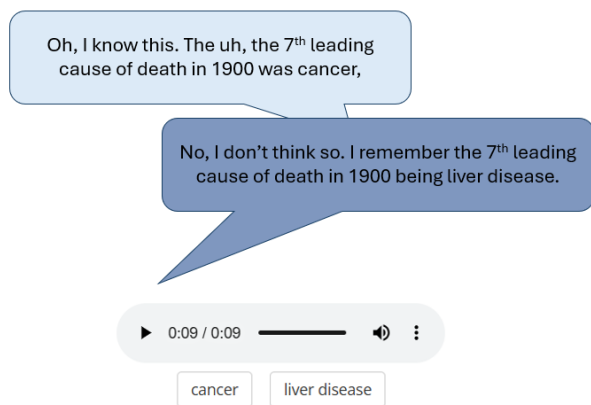
ing items where the two utterances were not distinguished by fluency. Responses to filler items were not included in the analysis.

2.2.2. Participants

A total of 48 participants (28 female, 20 male) were recruited via Prolific² to take part in a web-based listening task on cognition.run³. Participants were native speakers of English from the United States, UK, Ireland, Canada and Australia, with a mean age of 41.2. They were pre-screened to ensure that they had not participated in the pilot study and had a submission approval rate of at least 90% on Prolific.

2.2.3. Listening task

Participants were informed that the correct answer might not be among the two alternatives, and told to pick the option they thought was closest to the correct answer in such cases. They were also made aware that the voices were computer generated.



Listen to the audio and select the correct answer.

Figure 1: An experimental trial in the listening task. Note: the blue speech bubbles are included to represent what the participants heard, but no transcripts of the audio were shown in the actual experiment

In order to incentivize efforts to do well at the task, we told participants that bonus payments would be awarded based on performance. In reality, this threshold was extremely low. All participants who completed the experiment and passed the two attention checks were given a 33% bonus. Given that none of the answers presented were actually correct, it would have been impossible to award performance beyond this minimal criterion. Furthermore, the incentive was intended to encourage sincere effort at being accurate rather than to reward specific choices during the task.

Before the experimental trials, participants completed a brief practice round to ensure they understood the format of the experiment. The stimuli played automatically at the start of each trial, but participants were able to play the audio again if needed. However, they were encouraged not to spend too much time on each answer. Participants responded by clicking the button beneath the audio player that corresponded to what they thought was the correct answer. The buttons were enabled only after the audio had finished playing the first time.

²<https://www.prolific.com/>

³<https://www.cognition.run/>

After the experimental trials, participants were asked to rate their overall level of confidence in the answers they had given, on a scale ranging from 1 (very unsure) to 5 (very confident). This item was optional.

3. Results

3.1. Effects of fluency on answer choice

The percentage of disfluent vs. fluent answers chosen by participants as correct is shown in Figure 2. We conducted a one proportion Z test to determine whether the proportion of fluent answers chosen as correct significantly differed from chance (50%). The test showed that fluent answers were significantly more likely to be chosen as correct than disfluent answers, $Z = 2.53$, p (two-tailed) < 0.05 .

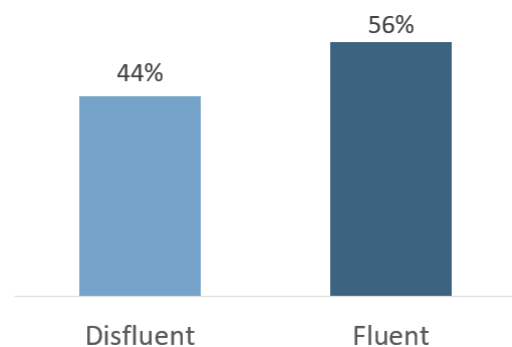


Figure 2: Percentage of answers chosen as correct by fluency

3.2. Confound checks

In addition to the extensive counterbalancing described in Section 2.1.6, we carried out several checks to further rule out confounds related to the speaker's voice or a bias toward selecting the first vs. the second utterance as correct.

We tested whether listeners chose any particular voice as correct more often than would be expected by carrying out a Pearson's χ^2 . The distribution of "correct" responses across speakers did not differ significantly from chance, $\chi^2(3) = 1.77$, $p = 0.62$.

Effects of utterance order were tested with a one proportion Z test. Participants did not show a preference for selecting the first utterance as the correct one, or vice versa, $Z = 0.61$, p (two-tailed) $= 0.54$.

3.3. Participants' confidence in their performance

Since the confidence rating was optional, not every participant completed it. Out of 48 participants, 34 responded to this item. Their mean rating on the 5-point scale was 2.65 (SD=1.08).

4. Discussion

Our results suggest that listeners take disfluency into account when deciding between conflicting sources of information, in the context of a presumed reward for making the right decisions. This expands upon previous findings that speech disfluencies guide decisions about whether to trust a potentially dishonest speaker [16, 12], showing that listeners also use these cues when deciding who or what to believe, even in the absence of any clear expectation of deliberate misinformation.

While our task was less naturalistic than the social games used by [16] and [12], it was also significantly easier and less costly to implement. Furthermore, while a number of factors were not controlled for in our experiment (for example, prosodic characteristics of the voices or the length of utterances), we were able to rule out ordering effects and blanket effects of simply preferring one voice over another.

We also believe that our findings are potentially more informative about how listeners use the information provided by speech disfluencies than rating scale responses. Rating scales are a good starting point for establishing whether an association exists between some aspect of speech and listener judgments, but more behavioral tasks are needed to understand the practical implications of these associations. Experiments such as ours could be used to test hypotheses about how speech characteristics influence behavior, and then follow up on promising results with larger-scale studies. This type of task also lends itself well to combining with online measurement techniques such as electroencephalography (EEG). EEG can provide valuable insights into the cognitive processes underlying decision-making, but using EEG recordings in conversational settings poses substantial methodological challenges [18].

One potential criticism of our study is that we asked participants to make a choice in which neither option was correct. Participants' fairly lukewarm assessment of how well they performed at the task seems to indicate that they found it challenging. This could have magnified the effects of any salient differences between stimuli and forced listeners to rely more heavily on cues that might not be as influential in real-life decision-making.

While being forced to make decisions without much to go on is by no means an unusual situation, future work should explore how disfluencies affect decisions where the choices are less ambiguous. Can disfluency cause listeners to doubt information that they initially believe is correct? This could be investigated by manipulating the objective reliability of a speaker (for example, by providing several rounds of feedback about the accuracy of the answers they give) or using questions that do have a correct answer.

Since we used synthesized speech to create our stimuli, and informed participants of this, one might argue that listeners could not meaningfully make the kinds of inferences about a speaker's knowledge that are needed to decide who to believe. However, there is evidence that listeners do make such inferences even when they are fully aware that the "speaker" is not a human. Discounting effects that rely on metacognitive processes have been shown in previous experiments using synthetic speech, where the speaker admitting to being anxious made listeners more forgiving of their disfluency despite knowing that they were not a real person [6]. Nonetheless, future work could delve more deeply into how listeners form these kinds of inferences when interacting with non-human conversation partners such as social robots.

Finally, our experiments used the same types of disfluencies in roughly the same locations in all of the disfluent utterances. In light of evidence that the type and position of speech disfluencies affect their degree of impact on judgments [8, 5, 6], it would be interesting to explore whether these same factors play a role in decision-making.

5. Conclusions

Our findings suggest that speech disfluencies, in addition to impacting evaluations of a speaker's competence or confidence,

can influence motivated decision-making. Listeners who were incentivized to perform well at a task chose answers that were delivered fluently at a higher rate than disfluent answers. Future work should look into whether fluency can sway listeners when the choices are less ambiguous than the ones we presented. Future studies should also consider whether the type and location of disfluencies affect decisions to the same extent that they affect judgments.

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